REACTIVEDYNAMICS.JL — THE AGENTIC LAYER

ReactiveDynamics.jl Modeling Copilot

Describe the operation in conversation. The copilot builds a high-fidelity model beneath it, one that carries the shared capacity, lead times, feedback and mid-course decisions a spreadsheet flattens, tries to break it, and uses it to compare the options and explain what drives the answer. The conversation stays in the terms of the business, never the model's mechanics; the model follows from what is said.

THE PROBLEM

Decisions about how an operation behaves need a model, and a model needs a modeler

- A launch reservation, a bid under a staffing ceiling, a pipeline whose funding follows its results, a backlog that feeds back into demand: the answer depends on **how the operation behaves over time**.
- A workbook can hold a time series and even a Monte Carlo draw. It cannot easily hold one shared line allocated month by month between products with random launch dates, plus a rule that changes the plan when a readout lands. That takes a simulator nobody can check.
- So these questions go to a specialist, take weeks, and come back as a workbook the decision-maker has to take on faith.
- Most never get a model at all.

More decisions get a real model, the owner stays in the loop, the answer travels

- **A model for the cost of a conversation.** The threshold for a real model drops from a specialist's weeks to a guided session with the decision owner.
- **The owner stays in the loop.** The interview runs in business language with the person who owns the decision. Every assumption lands in a brief they signed, including which inputs were their own estimates.
- **The answer travels.** A study ends in a generated report the modeler can forward: the answer, the model, the evidence and the provenance, laid out to one template.
- **It compounds.** The next question about the same operation starts from the same model.

THE REACH

What it can model — and what the copilot checks today

Pattern	Example	Copilot today
Contention	two launches on one fill line	authored and checked end to end
Lead times, clocks	tech transfer, a build, a review	authored and checked
Feedback	backlog throttles intake; share lost after a stock-out	authored and run (12 state-dependent rates in the case); loop checks next (W25)
Mid-course decisions	review a contract at a readout	authored and run; the case turned on one
Lifecycles	projects moving through phases	authored and run
Coupled models	supply feeding commercial; several sites	in the engine (AlgebraicAgents); not yet through the copilot (W25)

HOW IT WORKS

Four steps, and a check that can stop the answer

From a question to a number you can take to a board

Four steps, and the third one is allowed to stop the other three.

01

Describe the decision

A conversation: the choice, what drives it, how the parts interact. If arithmetic answers it, we stop here.

02

We build your operation

A running model of how it behaves — shared capacity, delays, feedback, decisions taken mid-course, cash.

03

We try to break it

Nine checks ask whether the model does what you said the business does. A failure blocks the answer.

04

A number you can defend

A central estimate with the evidence behind it — and a seed that reproduces it exactly.

If the model does not behave like your business, you get the diagnosis, not a number.

It names the check that failed and what it means: fix the model, or revisit the brief.

The model is data, not code, so it can be checked before it runs

Static checks. A pure validator reads the model document and returns every problem with its exact location. A malformed model produces diagnostics, not a plausible wrong number. The language model proposes, a deterministic checker disposes, and the loop repeats until the checker is silent.

The model is never code. It is a closed vocabulary the engine compiles; nothing in a document can name a program.

Where it stops today. In the case, the odds-weighting behind the headline ran in session-written scripts: published, re-run independently, but reviewed by reading. Moving it into `rdx effect` is W27.

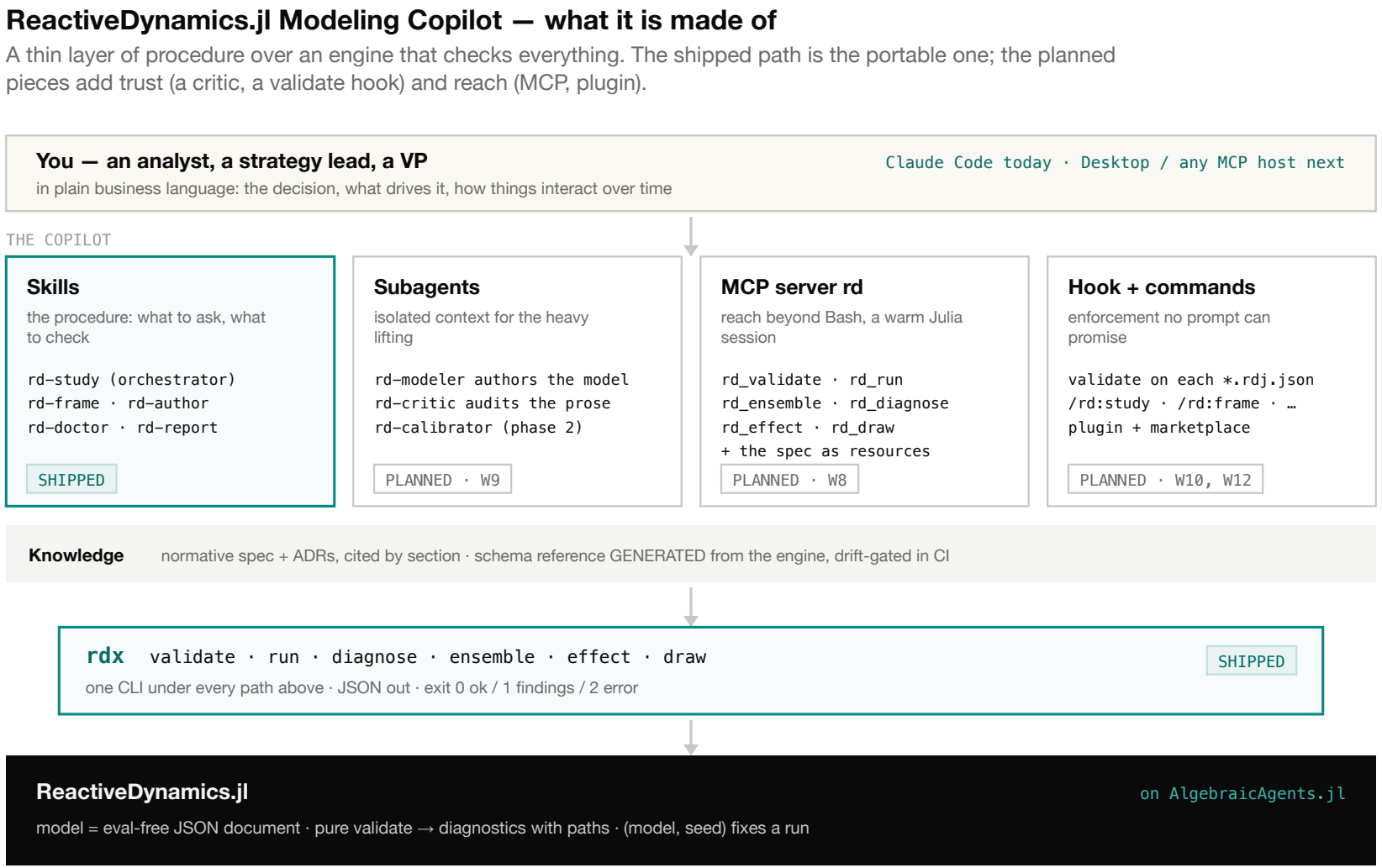
The model document. Places that hold resources, transitions that hold them over time and sometimes fail, rates that read the live state, rules that decide mid-course, costs, what each option changes. Diffable, and replayable from the model and a seed.

Checked again, for meaning. Plausibility checks ask whether the model does what the business does: a bottleneck that never binds, a step that never fires, a horizon that cuts off the payoff.

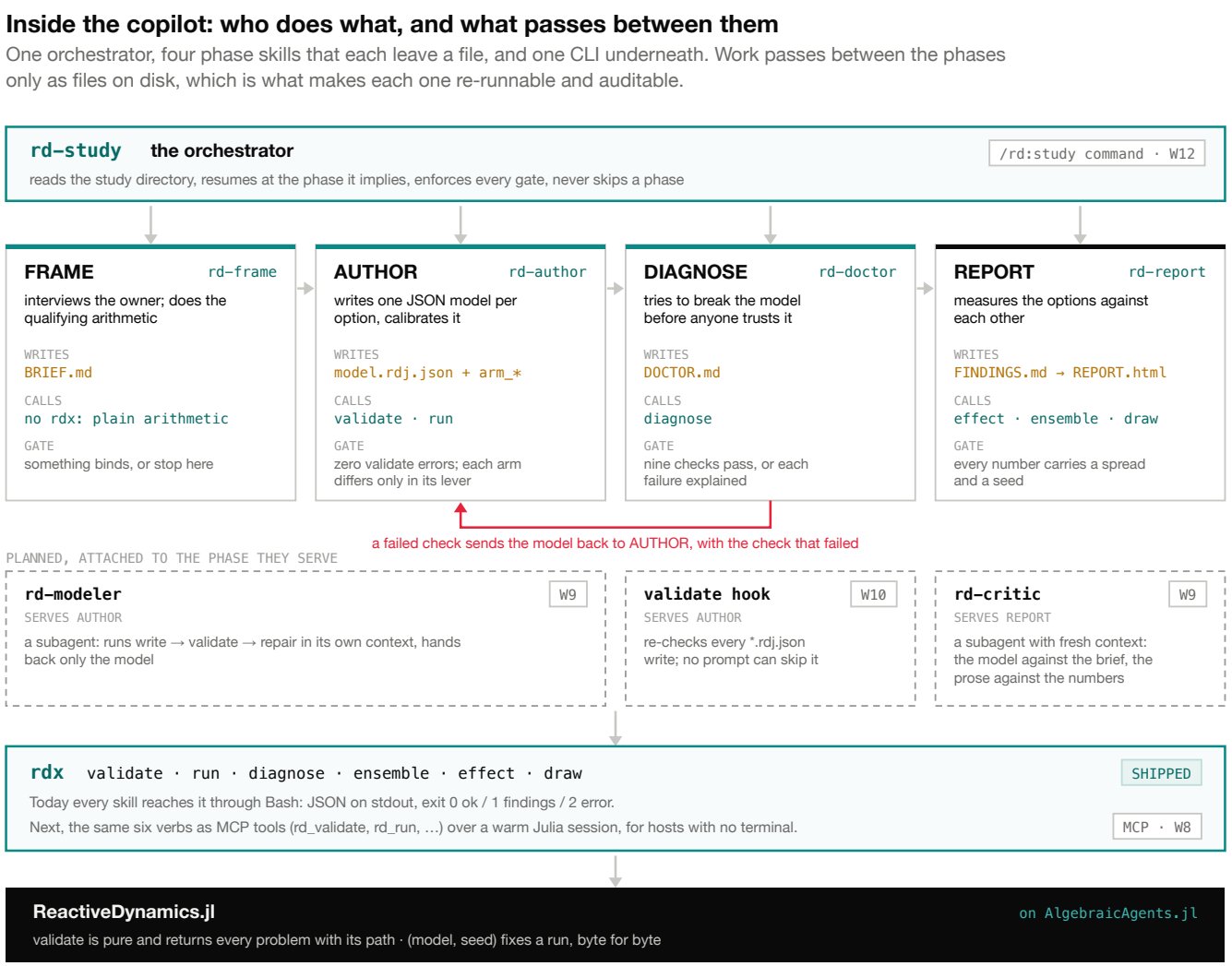
Guarded in CI. The schema reference is generated from the engine, and a known-answer test re-derives a published result on every change.

WHAT IT IS MADE OF

A thin layer over an engine that checks everything



Who does what, and what passes between them



A CAPTURED SESSION

Launch supply: do we pay for capacity before the readouts?

Both sides were AI: the operator was a role-played agent answering from a fact sheet sealed before the first message; the assistant never saw it. The company is invented; the numbers are real runs of the model.

Two Phase-3 biologics, one fill-finish line, a CMO offer that expires 30 November

Kestrel reads out first, at 60% odds; **Tern** nine months later, at 45%. Both would need the same line.

OPERATOR

Two Phase-3 biologics are coming. […] We have one sterile fill-finish line in-house, and most of it is already taken by our marketed products. It can probably carry one launch but not two. […] Our CFO says we shouldn't pay for capacity on an asset that has less than even odds.

CLAUDE

The CMO slot works for Tern's filing, but only just. If you sign by 30 November, PPQ is done somewhere between late 2027 and early 2028. […] So the CMO can be Tern's filed site with no PAS, but only if the CMO runs on time and Tern doesn't read out early.

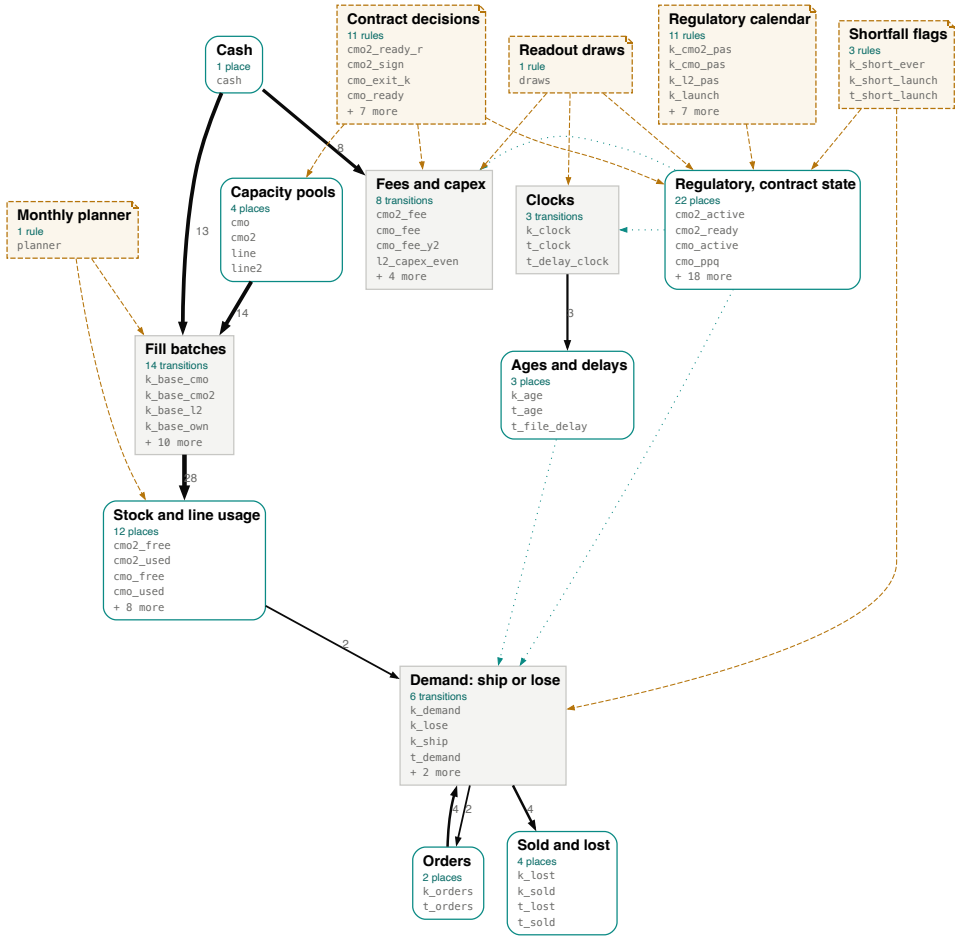
```
python3 -c '<in-house headroom: 12.8 batches/mo minus marketed 9/mo growing 5%/yr>'
```

One model, a version per option, checked against the operator's own figures

- A generator script writes one network; the options differ only in parameters.
- `rdx validate`: 0 errors on every document.
- Consistency check: Kestrel first-year revenue **$250.0M** against the operator's ~$250M, and **54** batches a year at peak. Both follow from the inputs by construction; they show the inputs went in right.
- `rdx diagnose`: the own line binds. Every option fails one check by design: one network holds every option and readout outcome, so a single run leaves the other paths' steps idle. `DOCTOR.md` shows each fires where its path is taken, before any number is reported.
- Options compared on **paired paths**: the same seed gives every option the same readout dates and lead times.
- Readout outcomes are forced per path and weighted by the stated odds (0.27 / 0.33 / 0.18 / 0.22).
- No option-level ensembles were run: a 96-seed ensemble took 53 minutes (W24). One wait ensemble served as a cross-check, and replays of three of its members matched exactly.

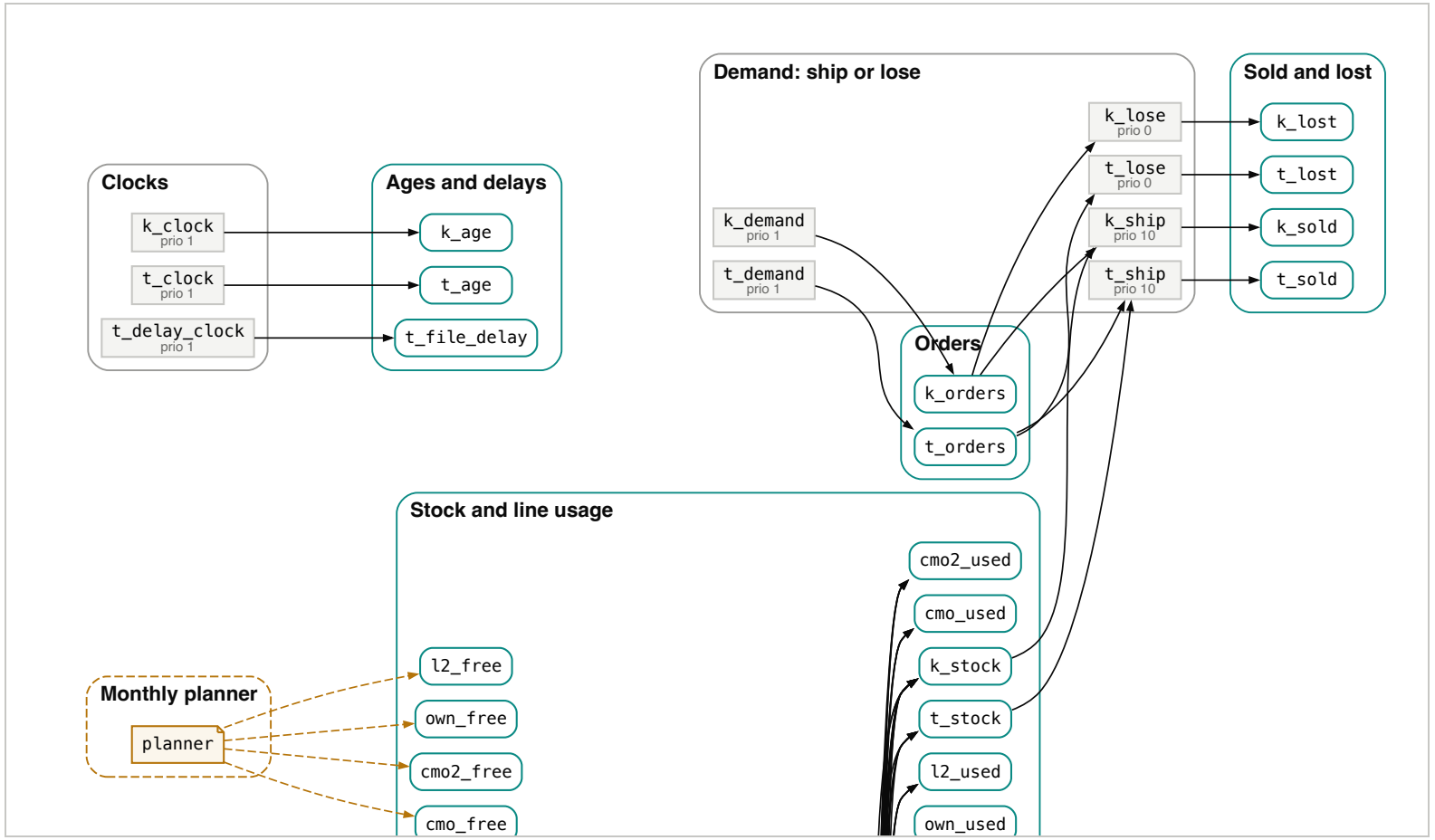
Every sentence the operator said is an element of the model

- *“the fill site has to be in the BLA with three PPQ batches before we file”* → rule `t_file`: Tern files only once a site has 30 tenths of PPQ.
- *“I'd move Tern's PPQ in-house … and give the CMO notice”* → rule `cmo_exit_k` and one switch, `exit_at_k = 1`: the only line that differs from the rigid reservation.
- The map is generated from the document: all 48 places, 31 transitions and 27 rules, each in one box.



The same model, every element

- **Places** (teal outline) are the stocks and pools: line capacity, stock, orders, cash.
- **Transitions** (grey) do the work, with their success chance and priority: `k_base_own` p 0.97 · prio 100.
- A **teal arc** is capacity returned when the batch is done; a black arc is consumed or produced; the number is how much.
- **Rules** (amber) are the calendar and the decisions, dashed where they set a place.
- All 48 places, 31 transitions and 27 rules. Nothing is summarized away.



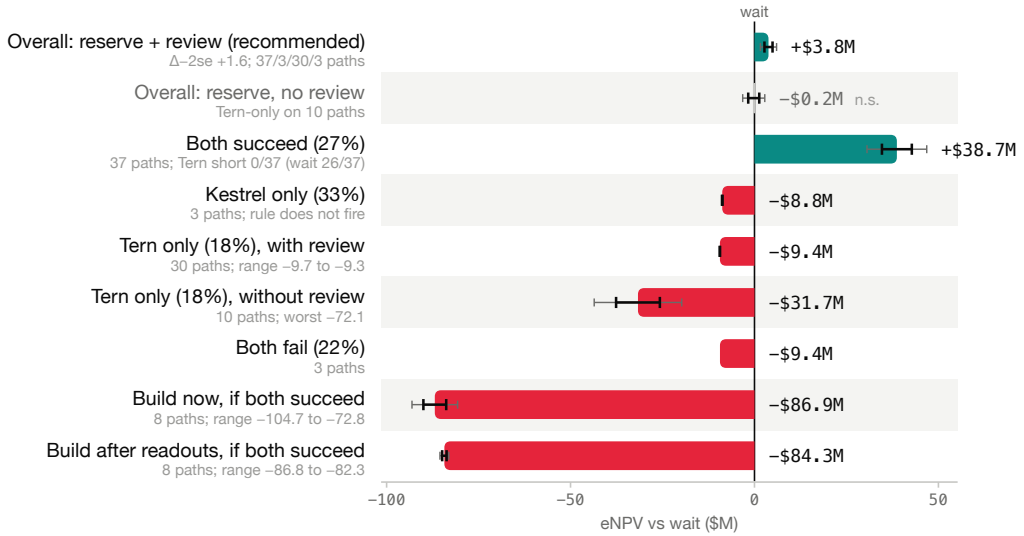
Scroll the diagram. Drawn from the engine's own view of the net (`network_graph`), inside the boxes of the map on the previous slide.

Reserve now, review at Kestrel's readout, don't build

- **+$3.8M ± $1.1M** against waiting, at the portfolio committee's odds; +$25M at peak-year batch value.
- **At the operator's own 50% for Kestrel: break-even** (+$1.6M ± $0.9M). This is supply protection, not a bet on the odds.
- **+$39M if both succeed** (27%): Tern short at launch on 26 of 37 paths under wait, 0 of 37 reserved.
- **A bounded $9–11M cost** in every other outcome.
- **Building is $73–105M worse** even when both succeed.
- **Tern-only is safe only with a second CMO contract in writing** (–$72.5M without, 3 paths, indicative).

Reserve the CMO now, review at Kestrel's readout: eNPV against waiting

Paired single paths (same seed, same dates in both options), whiskers ±1 se (dark) and ±2 se (grey) across paths. Top two bars: odds-weighted overall. Others: given the outcome. Kestrel 60%, Tern 45%, 9% WACC, year-1 value per batch.



The study ships as a generated report, not a folder

Six parts, one template, every study:

1. The answer and the decision rule
2. The evidence, sensitivities and caveats
3. Why a spreadsheet cannot produce it
4. The model: elements in the operator's words, and the diff that *is* the option
5. The execution map
6. How we got it: commands, commit, the seeds

Generated, never edited. `render_report.jl` builds it from the study's own files, so it cannot drift from them. Its prose is the study's own FINDINGS, written by the model; a critic that audits prose against numbers is W9.

Self-contained. One HTML file with a PDF beside it; every figure embedded, nothing to open in the repository.

Checkable. It names the commit, hashes both model documents, and lists all 208 stored runs by document and seed: any number can be replayed.

`agent/studies/launch_supply/REPORT.pdf`, 18 pages.

THE MOMENT WORTH WATCHING

The model made the tail visible, and the owner supplied the option

CLAUDE

In the other outcomes the reservation costs money: Tern only, 0.18 chance, **–$18.6M to –$72.1M**. […] That is why the expected value nets to about zero.

OPERATOR

Kestrel reads out in March 2027, nine months before Tern. […] I'd move Tern's PPQ in-house, file Tern on our own line, and give the CMO notice. That means eating the $5–10M. So the reservation isn't a fixed bet on Tern; it's something we decide to keep or drop at Kestrel's readout.

CLAUDE

Your guess holds: under the review rule, the Tern-only outcome costs $9.4M against waiting, and the $72M tail is gone.

```
python3 exit_review.py → weighted delta +3.81, se 1.10 (30 Tern-only paths)
```

HONEST ABOUT IT

What we are not claiming

- **The case is simulated, AI on both sides.** The operator was cooperative. It demonstrates the workflow, not anyone's supply plan.
- **Paired paths, not ensembles.** A 96-seed ensemble took 53 minutes (W24); every figure says what it rests on.
- **The interview left soft inputs untested:** demand ±30%, marketed growth and readout correlation. The published audit leads with them.
- **The prose is not yet checked.** The assistant retracted two of its own claims out loud; a critic that does this systematically is W9.
- **Contention is checked end to end; feedback and coupling are not yet** (W25).
- **The outlined pieces are the direction, not shipped.**

WHERE IT IS GOING

Trust first, then reach

Direction	What it changes	Tracker
Fast ensembles	an option-level ensemble in minutes, not 53	W24
Declared dead steps	a failed check means a bug	W20
Weighting in the engine	the headline comes out of `rdx effect`	W27
A critic	the briefing's prose audited against the numbers	W9
Beyond contention	checks for feedback; coupled models as documents	W25
Validate hook	every model edit validated automatically	W10
MCP server · plugin	any assistant, no terminal; one-step install	W8 · W12

The first half makes the answer harder to get wrong; the second moves the copilot from something a developer sets up to something an analyst opens.